

## # Perturbation theory and the S-matrix

- Let us show how the path-integral method is used in the calculation of scattering processes.

The scattering of one particle on another is described, non-relativistically, by interaction through a potential  $V(x)$  (we change the notation in this section from  $q$  to  $x$ ).

- Since the expression for the transition amplitude is not exactly calculable, we resort, as usual, to perturbation theory. This is valid when the potential  $V(x)$  is small, or, more precisely, when the time integral of  $V(x,t)$  is small compared with  $\hbar$ .

In that case we may write

- $$\exp\left[\frac{-i}{\hbar} \int_{t_i}^{t_f} V(x,t) dt\right] = 1 - \frac{i}{\hbar} \int_{t_i}^{t_f} V(x,t) dt - \frac{1}{2! \hbar^2} \left[ \int_{t_i}^{t_f} V(x,t) dt \right]^2 + \dots \quad (312)$$

→ this is the perturbation expansion.

When substituted into the expression (311) for the propagator  $K(x_f t_f; x_i t_i) = \langle x_f t_f | x_i t_i \rangle$  we get a series expansion

$$K = K_0 + K_1 + K_2 + \dots, \quad (313)$$

the first term of which is the free propagator  $K_0$ :

$$\begin{aligned} K_0 &= N \int \left[ \exp\left(\frac{i}{\hbar} S'\right) \right] \mathcal{D}x \\ &= N \int \left[ \exp\left(\frac{i}{\hbar} \int \frac{1}{2} m \dot{x}^2 dt\right) \right] \mathcal{D}x \end{aligned} \quad (314)$$

$$\therefore \langle x_f t_f | x_i t_i \rangle = N \int \mathcal{D}x \exp\left[\frac{i}{\hbar} \int_{t_i}^{t_f} L(x, \dot{x}) dt\right] \quad (311)$$

$$= N \int \mathcal{D}x \exp\left[\frac{i}{\hbar} S'\right]$$

$$= N \int \mathcal{D}x \exp\left[\frac{i}{\hbar} \int_{t_i}^{t_f} \left[ \frac{1}{2} m \dot{x}^2 - V(x, t) \right] dt\right]$$

$$= N \int \mathcal{D}x \exp\left[\frac{i}{\hbar} \int_{t_i}^{t_f} \frac{1}{2} m \dot{x}^2 dt\right] \cdot \exp\left[-\frac{i}{\hbar} \int_{t_i}^{t_f} V(x, t) dt\right]$$

$$= N \int \mathcal{D}x \exp\left[\frac{i}{\hbar} \int_{t_i}^{t_f} dt \frac{1}{2} m \dot{x}^2\right] \cdot \left[ 1 - \frac{i}{\hbar} \int_{t_i}^{t_f} V(x, t) dt - \frac{1}{2! \hbar^2} \left[ \int_{t_i}^{t_f} V(x, t) dt \right]^2 + \dots \right] \quad (315)$$

To evaluate  $K_0$ , we write it in the discrete form (see (310))

$$K_0 = \lim_{m \rightarrow \infty} \left( \frac{m}{i\hbar\tau} \right)^{(m+1)/2} \underbrace{\int_{-\infty}^{\infty} \left( \prod_{j=1}^m dx_j \right) \exp \left[ \frac{im}{2\hbar\tau} \sum_{j=0}^m (x_{j+1} - x_j)^2 \right]}_I. \quad (316)$$

The value of  $I$  is known:

$$I = \frac{1}{(m+1)^{1/2}} \left( \frac{i\hbar\tau}{m} \right)^{m/2} \exp \left[ \frac{im}{2\hbar(m+1)\tau} (x_f - x_i)^2 \right], \quad (317)$$

so, putting  $(m+1)\tau = t_f - t_i$ , we have for the free particle propagator

$$K_0(x_f t_f; x_i t_i) = \left( \frac{m}{i\hbar(t_f - t_i)} \right)^{1/2} \exp \left[ \frac{im(x_f - x_i)^2}{2\hbar(t_f - t_i)} \right], \quad t_f > t_i. \quad (318)$$

The condition  $t_f > t_i$  is clearly crucial for the propagator, since it vanishes, by causality, if  $t_f < t_i$ , so we may properly put

$$K_0(x_f t_f; x_i t_i) = \theta(t_f - t_i) \left( \frac{m}{i\hbar(t_f - t_i)} \right)^{1/2} \exp \left[ \frac{im(x_f - x_i)^2}{2\hbar(t_f - t_i)} \right]. \quad (319)$$



PROOF of (317): we can show by induction that

$$\int_{-\infty}^{\infty} \exp \left\{ i\lambda \left[ (x_1 - a)^2 + (x_2 - x_1)^2 + \dots + (b - x_m)^2 \right] \right\} dx_1 \dots dx_m \\ = \left[ \frac{i^m \pi^m}{(m+1)\lambda^m} \right]^{1/2} \exp \left[ \frac{i\lambda}{m+1} (b-a)^2 \right]; \quad (320)$$

we assume it is true for  $m$ , and show it is true for  $m+1$ . We have

$$\int_{-\infty}^{\infty} \exp \left\{ i\lambda \left[ (x_1 - a)^2 + \dots + (b - x_{m+1})^2 \right] \right\} dx_1 \dots dx_{m+1} \\ = \left[ \frac{i^m \pi^m}{(m+1)\lambda^m} \right]^{1/2} \int_{-\infty}^{\infty} \exp \left[ \frac{i\lambda}{m+1} (x_{m+1} - a)^2 \right] \exp \left[ i\lambda (b - x_{m+1})^2 \right] dx_{m+1} \\ = \left[ \frac{i^m \pi^m}{(m+1)\lambda^m} \right]^{1/2} \int_{-\infty}^{\infty} \exp \left\{ i\lambda \left[ \frac{1}{m+1} (x_{m+1} - a)^2 + (b - x_{m+1})^2 \right] \right\} dx_{m+1}. \quad (321)$$

Putting  $x_{m+1} - a = y$ , the term in square brackets becomes

$$\frac{1}{m+1} y^2 + (b - a - y)^2 = \frac{m+2}{m+1} y^2 - 2y(b-a) + (b-a)^2 \\ = \frac{m+2}{m+1} \left[ y - \frac{m+1}{m+2} (b-a) \right]^2 + \frac{1}{m+2} (b-a)^2. \quad (322)$$

Now we put  $\lambda - (m+1/m+2)(b-a) = z$  and find that the integral is

$$\begin{aligned} & \left( \frac{i^m \pi^m}{(m+1)\lambda^m} \right)^{1/2} \int_{-\infty}^{\infty} \exp \left[ i\lambda \frac{m+2}{m+1} z^2 + \frac{i\lambda}{m+2} (b-a)^2 \right] dz \\ &= \left[ \frac{i^{m+1} \pi^{m+1}}{(m+2)\lambda^{m+1}} \right]^{1/2} \exp \left[ \frac{i\lambda}{m+2} (b-a)^2 \right] \end{aligned} \quad (323)$$

which is (320) with  $m+1$  instead of  $m$ . It only remains to show that the formula holds when  $m=1$ . In this case the integral is

$$\int_{-\infty}^{\infty} \exp \left\{ i\lambda \left[ \frac{3}{2} z^2 + \frac{1}{3} (b-a)^2 \right] \right\} dz = \left( \frac{2}{3} \frac{i\pi}{\lambda} \right)^{1/2} \exp \left[ \frac{i\lambda}{3} (b-a)^2 \right]. \quad (324)$$

This value for  $I$  is the same as what we get using the well-known integral

$$\int_{-\infty}^{\infty} \exp(-ax^2 + bx + c) dx = \exp \left( \frac{b^2}{4a} + c \right) \left( \frac{\pi}{a} \right)^{1/2}. \quad (325)$$

We have therefore proved (320) for all  $m$ .

Let us now calculate  $K_1$ . From (310) and (312) we have

$$K_1 = \frac{-i}{\hbar} \lim_{m \rightarrow \infty} N^{(m+1)/2} \sum_{i=1}^m \tau \int \exp \left[ \frac{im}{2\hbar\tau} \sum_{j=0}^m (x_{j+1} - x_j)^2 \right] V(x_i, t_i) dx_1 \dots dx_m, \quad (326)$$

where  $N = m/\hbar\tau$ , and we have replaced integration over  $t$

by summation over  $t_i$ .

$$K_1 = N \int dx \exp \left[ \frac{i}{\hbar} \int_{t_i}^{t_f} dt \frac{1}{2} m \dot{x}^2 \right] \left[ \frac{(-i)}{\hbar} \int_{t_i}^{t_f} dt V(x, t) \right]$$

$$\langle x_f t_f | x_i t_i \rangle = \lim_{m \rightarrow \infty} \left( \frac{m}{i\hbar\tau} \right)^{(m+1)/2} \int_{-\infty}^{\infty} \left( \frac{m}{\pi} dx_j \right) \exp \left\{ \frac{i\tau}{\hbar} \sum_{j=0}^m \left[ \frac{m}{2} \left( \frac{x_{j+1} - x_j}{\tau} \right)^2 - V \right] \right\}$$

$$= \lim_{m \rightarrow \infty} \left( \frac{m}{i\hbar\tau} \right)^{(m+1)/2} \int_{-\infty}^{\infty} \left( \frac{m}{\pi} dx_j \right) \exp \left\{ \sum_{j=0}^m \left[ \frac{i\tau m}{2\hbar} \frac{(x_{j+1} - x_j)^2}{\tau^2} - \frac{i\tau}{\hbar} V \right] \right\}$$

$$= \lim_{m \rightarrow \infty} \left( \frac{m}{i\hbar\tau} \right)^{(m+1)/2} \int_{-\infty}^{\infty} \left( \frac{m}{\pi} dx_j \right) \exp \left\{ \sum_{j=0}^m \frac{i\tau m}{2\hbar} \left( \frac{x_{j+1} - x_j}{\tau} \right)^2 \right\} \exp \left\{ \sum_{j=0}^m \frac{-i\tau V}{\hbar} \right\}$$

Using  $\exp \left[ \frac{-i}{\hbar} \int_{t_i}^{t_f} dt V(x, t) \right] = \lim_{m \rightarrow \infty} \exp \left[ \frac{-i\tau}{\hbar} \sum_{i=1}^m V(x_i, t_i) \right]$  with

$t_1 = t_i$  and  $t_m = t_f$ , we can find (326).

Noting that  $V$  depends on  $x_i$ , we now split up the sum in the exponent into two, one going from  $j=0$  to  $j=i-1$ , and the other from  $j=i$  to  $j=m$ . We also separate out the integration over  $x_i$ , and get:



$$\begin{aligned}
K_1 &= \lim_{m \rightarrow \infty} \frac{-i}{\hbar} N^{(m+1)/2} \sum_{i=1}^m \tau \int \dots \int \exp \left\{ \frac{im}{2\hbar\tau} \sum_{j=0}^{i-1} (x_{j+1} - x_j)^2 \right\} \\
&\quad \times \exp \left\{ \frac{im}{2\hbar\tau} \sum_{j=i}^m (x_{j+1} - x_j)^2 \right\} V(x_i, t_i) dx_1 \dots dx_m \\
&= \lim_{m \rightarrow \infty} \frac{-i}{\hbar} \sum_{i=1}^m \tau \int \dots \int dx_i \left\{ N^{\frac{(m-i+1)}{2}} \exp \left[ \frac{im}{2\hbar\tau} \sum_{j=i}^m (x_{j+1} - x_j)^2 \right] dx_{i+1} \dots dx_m \right\} \\
&\quad \times V(x_i, t_i) \underbrace{\left\{ N^{i/2} \exp \left[ \frac{im}{2\hbar\tau} \sum_{j=0}^{i-1} (x_{j+1} - x_j)^2 \right] dx_1 \dots dx_{i-1} \right\}}_B
\end{aligned} \tag{327}$$

The two terms in curly brackets are  $A = K_0(x_f t_f; x t)$  and  $K_0(x t; x_i t_i)$ , so, replacing  $\sum_{i=1}^m \tau \int dx_i$  by  $\int dx dt$ ,

the above expression becomes

$$K_1(x_f t_f; x_i t_i) = \frac{-i}{\hbar} \int_{t_i}^{t_f} dt \int_{-\infty}^{\infty} K_0(x_f t_f; x t) V(x, t) K_0(x t; x_i t_i) dx. \tag{328}$$

Now  $K_0(x_f t_f; x t)$  vanishes if  $t > t_f$  and  $K_0(x t; x_i t_i)$  vanishes if  $t < t_i$ , so the integration in (328) may be taken over all values of  $t$ , to give

$$K_1(x_f t_f; x_i t_i) = \frac{-i}{\hbar} \int_{-\infty}^{\infty} dt \int_{-\infty}^{\infty} K_0(x_f t_f; x t) V(x, t) K_0(x t; x_i t_i) dx. \tag{329}$$

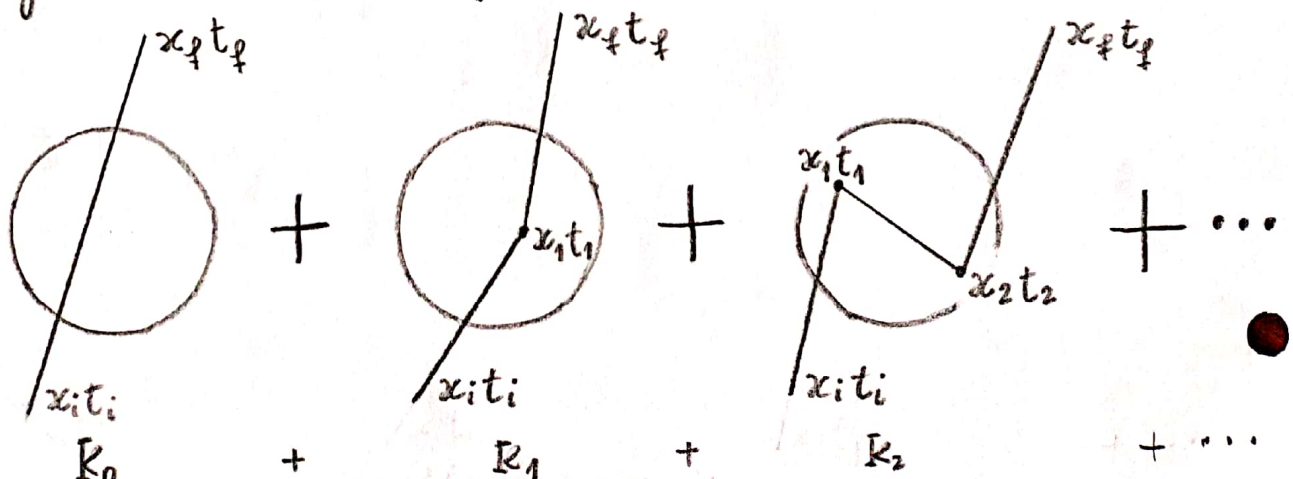
This is the first order correction to the free propagator. In a similar way, we may prove that the second order correction is

$$K_2(x_f t_f; x_i t_i) = \left(\frac{-i}{\hbar}\right)^2 \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 \int_{-\infty}^{\infty} dx_1 \int_{-\infty}^{\infty} dx_2 K_0(x_f t_f; x_2 t_2) V(x_2, t_2) \\ \times K_0(x_2 t_2; x_1 t_1) V(x_1, t_1) K_0(x_1 t_1; x_i t_i). \quad (330)$$

Analogous expressions hold for all  $K_m$  in the expansion (313), so we write

$$K(x_f t_f; x_i t_i) = K_0(x_f t_f; x_i t_i) - \frac{i}{\hbar} \int K_0(x_f t_f; x_1 t_1) V(x_1, t_1) K_0(x_1 t_1; x_i t_i) dx_1 dt_1 \\ - \frac{1}{\hbar^2} \int K_0(x_f t_f; x_1 t_1) V(x_1, t_1) K_0(x_1 t_1; x_2 t_2) V(x_2, t_2) \\ \times K_0(x_2 t_2; x_i t_i) dx_1 dx_2 dt_1 dt_2 + \dots \quad (331)$$

This is the solution to the perturbation series in  $K$ , and is called the Born series. It may be visualized as in this Figure:





- $K_0$  describes the free propagation of the wave function from  $x_i t_i$  to  $x_f t_f$ ;  $K_1$  describes propagation with one interaction with the potential  $V$ ; and so on.

A noteworthy feature of (330) is that it does not include the factor  $1/2!$  present in (312). The reason for this is as follows. The two interactions with  $V$  occur at different times but are indistinguishable, so we write

$$\begin{aligned} \frac{1}{2!} \int V(t') V(t'') dt' dt'' &= \frac{1}{2!} \int [\theta(t' - t'') V(t') V(t'') \\ &\quad + \theta(t'' - t') V(t') V(t'')] dt' dt'' \\ &= \int \theta(t_1 - t_2) V(t_1) V(t_2) dt_1 dt_2. \end{aligned} \quad (332)$$

- In a similar way, there is no factor  $1/n!$  in the expression for  $K_n$ .
- We shall now show that the free propagator  $K_0$  is simply the Green's function for the Schrödinger equation. To see this, substitute the Born series (331) into  $\Psi(x_f t_f) = \int K(x_f t_f; x_i t_i) \Psi(x_i t_i) dx_i$ , giving

$$\begin{aligned}
\psi(x_f t_f) &= \int K(x_f t_f; x_i t_i) \psi(x_i t_i) dx_i \\
&= \int K_0(x_f t_f; x_i t_i) \psi(x_i t_i) dx_i \\
&\quad - \frac{i}{\hbar} \int K_0(x_f t_f; x t) V(x, t) K_0(x t; x_i t_i) \psi(x_i t_i) dt dx dx_i \\
&\quad + \dots
\end{aligned} \tag{333}$$

→ here we have changed from one space dimension to three.

Assuming the series above converges, the effect of the unwritten terms is to modify the last  $K_0$  to the full propagator, so that

$$\psi(x_f t_f) = \int K_0(x_f t_f; x_i t_i) \psi(x_i t_i) dx_i - \frac{i}{\hbar} \int K_0(x_f t_f; x t) V(x, t) \psi(x, t) dx dt. \tag{334}$$

This equation is exact, and is an integral equation for  $\psi$ .

Now assume that in the distance past,  $t_i \rightarrow -\infty$ ,  $\psi$  becomes free — a plane wave  $\phi$  — then the first term on the right-hand side of (334) is also a plane wave, since it results from the free propagation of  $\psi(x_i t_i)$ , and we may write

$$\psi(x_f t_f) = \phi(x_f t_f) - \frac{i}{\hbar} \int K_0(x_f t_f; x t) V(x, t) \psi(x, t) dx dt. \tag{335}$$



Now  $\Psi(x_f t_f)$  obeys the Schrödinger equation

$$\frac{\hbar^2}{2m} \nabla_{x_f}^2 \Psi(x_f t_f) + i\hbar \frac{\partial \Psi(x_f t_f)}{\partial t_f} = V(x_f t_f) \Psi(x_f t_f). \quad (336)$$

Since  $\phi(x_f t_f)$  obeys the free-particle equation (with  $V=0$ ), then  $K_0$  must obey

$$\frac{\hbar^2}{2m} \nabla_{x_f}^2 K_0(x_f t_f; x t) + i\hbar \frac{\partial}{\partial t_f} K_0(x_f t_f; x t) = i\hbar \delta(x_f - x) \delta(t_f - t) \quad (337)$$

which is the equation for the Green's function of (336).

Note that the presence of the  $\delta(t_f - t)$  is to be expected from the  $\theta(t_f - t)$  occurring in the definition (319) of  $K_0$ . The propagator  $K_0$  is then simply the

Green's function of the Schrödinger equation, as claimed.

# We pass now to the calculation of the scattering amplitude. In the measurement of a scattering process, the experimental conditions are that a particle is free at  $t = -\infty$ , then scatters, and is free again at  $t = +\infty$ .



This is a source of difficulty, however, because a free particle (that is, one with a definite energy and momentum) is described by a plane wave, which spreads out over all space and time, including the centre of the interaction potential  $V(x)$ , so the particle can never be free!

To get round this problem the adiabatic hypothesis is invoked; the potential  $V$  is switched on and off again slowly, so that  $V=0$  at  $t=+\infty$  and the particle is free.  $V$  must not be switched on and off quickly; this would imply, through Fourier transformation, that the time dependence of  $V$  results in the scattering centre emitting or absorbing energy, which must not happen.

Returning to the scattering problem, the initial condition is that  $\Psi$  is a plane wave:

$$\Psi_{in}(x; t_i) \text{ plane wave.} \quad (338)$$

We assume that  $V \rightarrow 0$  for large negative  $t$ , and that  $t_i$  is even further in the past. Taking the first Born approximation in (333) gives

$$\begin{aligned} \Psi^{(+)}(x_f t_f) = & \int K_0(x_f t_f; x_i t_i) \Psi_{in}(x_i t_i) dx_i \\ & - \frac{i}{\hbar} \int K_0(x_f t_f; x t) V(x, t) K_0(x t; x_i t_i) \Psi_{in}(x_i t_i) dx dx_i dt. \end{aligned} \quad (339)$$

The superscript on  $\Psi^{(+)}(\mathbf{x}_f t_f)$  denotes that it corresponds to a wave which was free at  $t = -\infty$ , and thus involves the "retarded" propagator  $K_0(\mathbf{x}t; \mathbf{x}'t')$ , which vanishes for  $t' > t$ .

It is equally acceptable to write a solution  $\Psi^{(-)}(\mathbf{x}, t_i)$  consisting of a wave which becomes free at  $t = \infty$  ( $\Psi_{\text{OUT}}$ ), and an "advanced" propagator  $K_0(\mathbf{x}t; \mathbf{x}'t')$  which vanishes when  $t' < t$ .

We are interested in the amplitude for detecting a final particle with definite momentum, i.e. a plane wave  $\Psi_{\text{OUT}}$ . This is called the scattering amplitude  $S'$ , and is the overlap of the wave functions

$$S' = \int \Psi_{\text{OUT}}^*(\mathbf{x}_f t_f) \Psi^{(+)}(\mathbf{x}_f t_f) d\mathbf{x}_f$$

$$= \int \Psi_{\text{OUT}}^*(\mathbf{x}_f t_f) K_0(\mathbf{x}_f t_f; \mathbf{x}_i t_i) \Psi_{\text{in}}(\mathbf{x}_i t_i) d\mathbf{x}_i d\mathbf{x}_f$$

$$- \frac{i}{\hbar} \int \Psi_{\text{OUT}}^*(\mathbf{x}_f t_f) K_0(\mathbf{x}_f t_f; \mathbf{x}t) V(\mathbf{x}, t) K_0(\mathbf{x}t; \mathbf{x}_i t_i) \Psi_{\text{in}}(\mathbf{x}_i t_i) d\mathbf{x}_f d\mathbf{x} d\mathbf{x}_i dt$$

$$= \int \Psi_{\text{OUT}}^*(\mathbf{x}_f t_f) \phi(\mathbf{x}_f t_f) d\mathbf{x}_f - \frac{i}{\hbar} \int \Psi_{\text{OUT}}^*(\mathbf{x}_f t_f) K_0(\mathbf{x}_f t_f; \mathbf{x}t)$$

$$\times V(\mathbf{x}, t) K_0(\mathbf{x}t; \mathbf{x}_i t_i) \Psi_{\text{in}}(\mathbf{x}_i t_i) d\mathbf{x}_f d\mathbf{x} d\mathbf{x}_i dt \quad (340)$$

where  $\phi(\mathbf{x}_f t_f)$ , just like  $\Psi_{\text{in}}(\mathbf{x}_i t_i)$ , is a plane wave.



If the initial and final momenta are  $\vec{p}_i = \hbar \vec{k}_i$ ,  $\vec{p}_f = \hbar \vec{k}_f$ , we have, with box normalisation,

$$\psi_{in}(\mathbf{x}, t) = \frac{1}{\sqrt{\mathcal{V}}} \exp \left[ \frac{i}{\hbar} (\vec{p}_i \cdot \mathbf{x} - E_i t) \right], \quad (341)$$

$$\psi_{out}(\mathbf{x}, t) = \frac{1}{\sqrt{\mathcal{V}}} \exp \left[ \frac{i}{\hbar} (\vec{p}_f \cdot \mathbf{x} - E_f t) \right]$$

where  $E = p^2/2m$  and  $\mathcal{V}$  is the volume of the box, which of course is arbitrary.

Substituting (341) into the first term of (340), using

$$\int e^{i\vec{q} \cdot \mathbf{x}} d\mathbf{x} = (2\pi)^3 \delta(\vec{q}), \quad (342)$$

and putting, for convenience,  $\mathcal{V} = (2\pi)^3$ , we get

$$S_{fi} = \delta(\vec{k}_i - \vec{k}_f) - \frac{i}{\hbar} \int \psi_{out}^*(\mathbf{x}_f, t_f) K_0(\mathbf{x}_f, t_f; \mathbf{x}, t) V(\mathbf{x}, t) \\ \times K_0(\mathbf{x}, t; \mathbf{x}_i, t_i) \psi_{in}(\mathbf{x}_i, t_i) d\mathbf{x}_f d\mathbf{x} d\mathbf{x}_i dt. \quad (343)$$

The scattering amplitude is then seen to be one element of a matrix  $S$ , whose  $(fi)$  element appears above; this is the scattering matrix or  $S$  matrix.



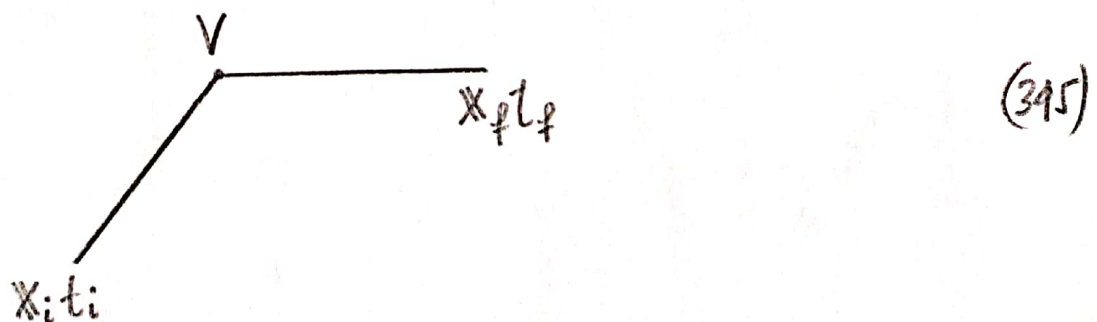
The first term corresponds to no interaction giving momentum conservation and a unit  $S$  matrix. Genuine interactions are represented by the second term in (343), and the amplitude that a particular "out" state results from a particular "in" state is

$$A = -\frac{i}{\hbar} \int \Psi_{out}^*(x_f, t_f) K_0(x_f, t_f; x, t) V(x, t) K_0(x, t; x_i, t_i) \times \Psi_{in}(x_i, t_i) dx_f dx dt. \quad (344)$$

We now have an expression for the scattering amplitude in terms of the free propagator  $K_0$  and the interaction potential  $V$ .

Equation (344) may be translated into a set of simple rules for the scattering amplitude; these rules are known as the Feynman rules.

We may represent amplitude (344) (which is a first order approximation), by the diagram



It is clear that the rules for translating this diagram into the expression for the scattering amplitude may be summarised by making the correspondence

$$\left. \begin{array}{l}
 \overline{x_1 t_1} \quad \overline{x_2 t_2} \quad K_0(x_2 t_2; x_1 t_1) \\
 \begin{array}{c} V \\ \diagup \quad \diagdown \\ x t \end{array} \\
 -\frac{i}{\hbar} V(x t); \text{ integration over } x \text{ and } t
 \end{array} \right\} \quad (346)$$

In addition, we multiply by  $\psi_{in}$  and  $\psi_{out}^*$  at the ends of the diagram and integrate over the two relevant spatial variables. Thus, the amplitude for the second order process

$$\begin{array}{c}
 x_i t_i \\
 \diagdown \\
 \begin{array}{c} V \\ \diagup \quad \diagdown \\ x t \end{array} \\
 \diagup \quad \diagdown \\
 \begin{array}{c} V \\ \diagup \quad \diagdown \\ x' t' \end{array} \\
 \diagup \quad \diagdown \\
 x_f t_f
 \end{array} \quad (347)$$

is

$$\begin{aligned}
 A^{(2)} = & \left( \frac{-i}{\hbar} \right)^2 \int \psi_{out}^*(x_f t_f) K_0(x_f t_f; x' t') V(x', t') K_0(x' t'; x t) V(x, t) \\
 & \times K_0(x t; x_i t_i) \psi_{in}(x_i t_i) dx_i dx dt dx' dt' dx_f. \quad (348)
 \end{aligned}$$

The rules (346) are called the Feynman rules.

● In the non-relativistic quantum mechanics we are dealing with at present, these rules are hardly necessary to do calculations, but in quantum field theory they are a great aid to calculation.